

RMSEA under FIML: Local Asymptotics with Missing Data

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Purpose

Reported FIML RMSEA reuses the complete-data formula verbatim: $\hat{\varepsilon} = \sqrt{\max(T - df, 0)/(n df)}$, with T the FIML likelihood-ratio statistic. Two things change underneath, and both matter.

1. **The population target moves with the design.** RMSEA estimates $\varepsilon = \sqrt{F_0/df}$, but under missing data F_0 is read in the *observed-information* metric, which averages over missingness patterns and is dominated by the complete-data information. The same structural misspecification therefore produces a *smaller* population RMSEA the more data are missing (Theorem 2). FIML RMSEA confounds misfit with information loss; it is not a pure model property.
2. **The robust correction needs the FIML sandwich.** The scaling/mixture that brings T back to a χ^2 reference is built from $U\Gamma_{\text{mis}}$ with $\Gamma_{\text{mis}} = V^{-1}JV^{-1}$, where the observed-data curvature V and the meat J both carry the pattern structure, and where (unlike complete data) the observed-versus-expected information choice is live even under normality.

This is the missing-data companion to the complete-data RMSEA note: the local-alternative scaffolding is identical, with the substitution $V \rightarrow$ the pattern-averaged FIML curvature and $\Gamma \rightarrow \Gamma_{\text{mis}}$. It also presupposes the point-estimate consistency of the FIML companion note (FIML targets η_0 only under the MAR conditional-moment conditions); if those fail, F_0 is not even the right population number to root.

1 The FIML statistic and its information objects

Row i has observed index set $o_i \subseteq \{1, \dots, p\}$. The saturated (H1) working model is Gaussian on the observed subvector,

$$\ell_i(\eta) = \log \phi_{|o_i|}\{x_{i,o_i}; \mu_{o_i}, \Sigma_{o_i o_i}\}, \quad \eta = (\mu, \text{vech } \Sigma) \in \mathbb{R}^{p^*}, \quad p^* = \frac{p(p+3)}{2}.$$

Missing data forces a mean structure, so η *always* carries the p means; $df = p^* - t$ counts them. Let $s_i = \partial \ell_i / \partial \eta$ be the casewise score (sparse in η : supported on the moments pattern o_i observes). Define the two saturated population objects

$$V = \text{plim } \frac{1}{N} \sum_i (-\partial^2 \ell_i / \partial \eta \partial \eta'), \quad J = \text{plim } \frac{1}{N} \sum_i \text{Cov}(s_i),$$

the observed-data normal *curvature* (expected information per case, averaged over the pattern distribution) and the score *meat*. The saturated FIML estimator obeys $\sqrt{N}(\hat{\eta}_{\text{sat}} - \eta_0) \xrightarrow{d} N(0, \Gamma_{\text{mis}})$ with the sandwich covariance $\Gamma_{\text{mis}} = V^{-1} J V^{-1}$. The structural model is $\theta \mapsto \sigma(\theta) \equiv \eta(\theta)$ with full-rank Jacobian $\dot{\sigma} = \partial \eta / \partial \theta'$, and the residual-information projector is, exactly as in the complete-data note,

$$U = V - V \dot{\sigma} (\dot{\sigma}' V \dot{\sigma})^{-1} \dot{\sigma}' V, \quad U \dot{\sigma} = 0, \quad \text{rank } U = df. \quad (1)$$

The FIML LRT is $T = 2\{\ell_{\text{sat}}(\hat{\eta}_{\text{sat}}) - \ell_{\text{model}}(\hat{\theta})\}$; a second-order expansion gives, with $w = \sqrt{N}(\hat{\eta}_{\text{sat}} - \eta(\theta^*))$,

$$T = w' U w + o_p(1). \quad (2)$$

Everything below is a statement about the quadratic form (2) with $w \xrightarrow{d} N(\tau, \Gamma_{\text{mis}})$.

2 Local asymptotics: the noncentral chi-square

The drift is the same device as in the complete-data note: a fixed off-model direction shrinking at the $\sqrt{N}$ rate, $\Sigma_N = \Sigma(\theta^*) + \Delta / \sqrt{N}$ (plus the matching mean drift), with limiting model error $\tau = \text{plim } w$ fixed.

Theorem 1 (Noncentral limit under FIML). *Under the drift, multivariate normality, MCAR, and identification (every η -coordinate seen by some pattern), the information identity $J = V$ holds, so $\Gamma_{\text{mis}} = V^{-1}$ and $U \Gamma_{\text{mis}} = U V^{-1}$ is idempotent of rank df . Hence*

$$T \xrightarrow{d} \chi_{df}^2(\lambda), \quad \lambda = \tau' U \tau = \lim_N N F_0.$$

Proof. Under MCAR + normality the saturated FIML score is a correctly specified likelihood score, so $J = V$ (information identity) and $\Gamma_{\text{mis}} = V^{-1}$. Then UV^{-1} is idempotent of rank df by the computation of the complete-data note (replace its V by the present curvature). A quadratic form $w'Uw$ with $w \sim N(\tau, V^{-1})$ and UV^{-1} idempotent of rank df is $\chi_{df}^2(\lambda)$, $\lambda = \tau'U\tau$. The population discrepancy expands in the same V -metric, giving $NF_0 \rightarrow \tau'U\tau$. $\square$

So the noncentrality interpretation survives: $\hat{\varepsilon} = \sqrt{\max(T - df, 0)/(n df)}$ estimates $\varepsilon = \sqrt{F_0/df}$. The difference from complete data is entirely in *which* V enters U , and that is where the design leaks in.

3 The target moves with the design

Write λ as a minimized quadratic form. From (1), $\tau'U\tau = \tau'V\tau - \tau'V\dot{\sigma}(\dot{\sigma}'V\dot{\sigma})^{-1}\dot{\sigma}'V\tau$, which is the V -orthogonal projection residual:

$$\lambda = \min_{b \in \mathbb{R}^t} (\tau - \dot{\sigma}b)' V (\tau - \dot{\sigma}b). \quad (3)$$

The structural error τ is a property of the population and the model, the same vector whatever is missing. Only V carries the missingness. Compare it to the complete-data curvature $V_c = \frac{1}{2}D'(\Sigma^{-1} \otimes \Sigma^{-1})D$ (bordered by the mean block).

Lemma 1 (Information monotonicity). *Each pattern's per-case information is the Fisher information of a coarsening (marginalization) of the complete vector, so $\mathcal{I}_{o_i} \preceq V_c$ in the Loewner order; averaging preserves it, $V = \text{plim } \frac{1}{N} \sum_i \mathcal{I}_{o_i} \preceq V_c$. Under identification $V \succ 0$.*

Theorem 2 (Missingness attenuates RMSEA). *Fix the structural misspecification τ . The FIML noncentrality is dominated by its complete-data value,*

$$\lambda_{\text{FIML}} = \min_b (\tau - \dot{\sigma}b)' V (\tau - \dot{\sigma}b) \leq \min_b (\tau - \dot{\sigma}b)' V_c (\tau - \dot{\sigma}b) = \lambda_{\text{complete}},$$

hence the population RMSEA satisfies $\varepsilon_{\text{FIML}} \leq \varepsilon_{\text{complete}}$, with the gap widening as the missing fraction grows.

Proof. By Lemma 1, $(\tau - \dot{\sigma}b)' V (\tau - \dot{\sigma}b) \leq (\tau - \dot{\sigma}b)' V_c (\tau - \dot{\sigma}b)$ for every b ; the minimum of a pointwise-smaller function is no larger. Divide by $n df$ and take roots. $\square$

Corollary 1 (FIML RMSEA is not design-invariant). *Two studies with identical population (μ_0, Σ_0) and identical model, but different missingness, report different population RMSEAs; a misfit that reads 0.08 complete will read smaller under attrition, not because the model fits better but because the missing data carry less information to expose the misfit. RMSEA cut-offs calibrated on complete data therefore drift conservative under missingness. A design-invariant rescaling would divide λ by the information fraction $\text{tr}(V)/\text{tr}(V_c)$ (a fraction-of-missing-information correction), recovering $\varepsilon_{\text{complete}}$; this is not what lavaan or magmaan report, and reporting it would require the complete-data curvature, which the incomplete sample does not deliver directly.*

The attenuation is small under light, conditionally-benign missingness and grows with the missing fraction; the empirical companion is the missing-data fit-index literature (Davey [4]).

4 Robust FIML RMSEA

Drop normality, or admit MAR/observed-information effects: the identity $J = V$ breaks and $\Gamma_{\text{mis}} = V^{-1}JV^{-1} \neq V^{-1}$. The expansion (2) stands, but $U\Gamma_{\text{mis}}$ is no longer idempotent.

Lemma 2 (General FIML limit). *Under the drift with finite fourth moments, $T \xrightarrow{d} \sum_{j=1}^{df} \alpha_j \chi_{1,j}^2(\delta_j)$, the α_j the nonzero eigenvalues of $U\Gamma_{\text{mis}}$ and $\sum_j \alpha_j \delta_j = \tau'U\tau$. Under $J = V$ all $\alpha_j = 1$ and the mixture collapses to Theorem 1.*

The exact-mean identity $\mathbb{E}[w'Uw] = \text{tr}(U\Gamma_{\text{mis}}) + \tau'U\tau$ again separates sampling noise from misspecification additively:

$$\mathbb{E}[T] \rightarrow \underbrace{\text{tr}(U\Gamma_{\text{mis}})}_{=c\,df} + \underbrace{\tau'U\tau}_{=NF_0}, \quad c = \frac{\text{tr}(U\Gamma_{\text{mis}})}{df} = \frac{\text{tr}(UV^{-1}JV^{-1})}{df}. \quad (4)$$

Here c is the FIML Satorra–Bentler/Yuan–Bentler (MLR) scaling: $c = 1$ under $J = V$, and $c \neq 1$ under non-normal or MAR-distorted scores. The misspecification term $\tau'U\tau = NF_0$ is built from U (hence V) alone and does not see J , so it is the *same* quadratic form as in Theorem 1. The robust estimator follows by subtracting the inflated mean:

$$\widehat{\varepsilon}_{\text{rob}} = \sqrt{\frac{\max(T - \hat{c}\,df, 0)}{n\,df}} = \sqrt{\hat{c}}\,\widehat{\varepsilon}_{\text{sc}}, \quad \hat{c} = \frac{\text{tr}(U\widehat{\Gamma}_{\text{mis}})}{df}, \quad (5)$$

with $\widehat{\Gamma}_{\text{mis}} = \hat{V}^{-1} \hat{J} \hat{V}^{-1}$ from the sample casewise scores and curvature (Yuan–Bentler [9]; the robust-RMSEA construction of Brosseau-Liard–Savalei [2] and Savalei [6]). As in the complete-data note, $\hat{\varepsilon}_{\text{rob}} = \sqrt{\hat{c}} \hat{\varepsilon}_{\text{sc}}$, and the scaled version (which divides the noncentrality by $\hat{c}$ a second time) over-corrects; (5) is the one that targets ε .

Remark (Observed vs. expected information: a FIML-only wrinkle). With complete data and a correct normal model, the curvature V is unambiguous. Under missing data the *observed* information (the realized casewise Hessians) and the *expected* information (their model expectation) differ in finite samples even when $J = V$ asymptotically, and they diverge further under non-normality (Savalei [5]). The choice enters $\hat{V}$ in (5), hence $\hat{c}$, hence the robust RMSEA and its interval. lavaan matches its MLR/robust output only under the observed-information, structured-H1 default; the FMG spectrum convention reads the eigenvalues of $U\Gamma_{\text{mis}}$ off the *unstructured* saturated H1. The point estimate is robust to the choice through the mean identity (4); the interval is not, and should declare its convention.

Remark (Spectrum vs. mean: where the interval comes from). The robust *point* estimate rests only on (4) and is correct to leading order regardless of the α_j . The interval inverts a noncentral χ^2_{df} CDF, approximating the true mixture of Lemma 2 by a single noncentral χ^2 ; the mean-and-variance (Satterthwaite) refinement replaces df by $df^* = \text{tr}(U\Gamma_{\text{mis}})^2 / \text{tr}\{(U\Gamma_{\text{mis}})^2\}$. The FMG/pEBA route used in the companion experiments instead keeps the full eigenvalue spectrum of $U\Gamma_{\text{mis}}$ and forms the exact mixture reference, which is the right object when a few α_j dominate and the single- χ^2 approximation is poor. magmaan should expose $\hat{c}$ (point), df^* (Satterthwaite interval), and the $U\Gamma_{\text{mis}}$ spectrum (pEBA interval) from one FIML information pass.

Remark (Truncation bites harder under FIML). Because $\lambda_{\text{FIML}} \leq \lambda_{\text{complete}}$ (Theorem 2), the true noncentrality is smaller under missingness, so $T - df$ (or $T - \hat{c} df$) lands below zero more often and $\hat{\varepsilon}$ piles up at the 0 floor. The negative- $\widehat{\text{n}\hat{\text{c}}\hat{\text{p}}}$ events seen in the experiment-23/24 cells are this finite-sample truncation artifact, not evidence of inconsistency; they should shrink with N and with the observed information, and the interval (Corollary above), not the truncated point estimate, is the honest summary.

Remark (Two-stage and the completion trap). The two-stage (ML2S) statistic first fills the saturated $\hat{\eta}_{\text{sat}}$ then fits the structure to it as if complete, so its naive LRT uses the *complete-data* curvature V_c in place of V , inflating T by roughly the information ratio (a $\approx 1.8\times$ over-rejection in the experiment-24 MCAR-normal cell). Feeding that naive T into the RMSEA formula mislabels the inflation as misfit. The Satorra–Bentler correction with the *correct*

two-stage sandwich (the Unstructured-H1 weight) restores the χ^2_{df} mean and reproduces lavaan `robust.two.stage`; the robust RMSEA (5) then transfers unchanged, with $\hat{V}$ the two-stage curvature.

Remark (Precondition: the target must exist). All of the above roots F_0 as the population FIML discrepancy. Under MAR this is the right target only when the incomplete-on-observed conditional means are linear and conditional covariances homoscedastic (the FIML consistency note); if those fail, $\hat{\eta} \not\rightarrow \eta_0$, F_0 does not measure structural misfit, and a robust correction repairs the reference law of a statistic that is no longer centered on the model. No scaling fixes a moved target; that is an estimator-bias problem, upstream of RMSEA.

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